





GlassBox: Personalized Calibration for Privacy-Preserving, Explainable Exam Proctoring

- Shama Begum
- Durgesh
- Arut Kumaran

GUIDE

MS. Mohana Priya

DEPARTMENT OF AIML

ABSTRACT

WHY IT MATTERS

Online exams have grown, and so has cheating. Facial recognition tools flag every student with one fixed rule unfair to darker skinned and disabled students.

WHAT WE BUILT

A browser based system using MediaPipe for gaze tracking, plus tab switching, focus loss, and copy paste signals. We fine tune RetinaFace on diverse skin toned data (FairFace) and run it in browser alongside MediaPipe to reduce bias.

HOW IT SCORES

Each student is calibrated to their own baseline in the first 20-30 seconds. A logistic regression classifier, trained on MSU OEP, gives teachers a green/yellow/red risk score disabled students get shifted thresholds.

VS. EXISTING TOOLS

Adds diversity aware detection, personalized calibration, and measured fairness not found in Proctorio or Examity.

LIMITATION

Browser only signals can't detect an undetected second device.

INTRODUCTION

The Shift

- Online exams have become a permanent fixture of education, accelerated by remote and hybrid learning
- Most existing proctoring tools rely on facial recognition to flag suspicious behavior
- These tools apply one fixed detection standard to every student, regardless of individual difference

The Fairness Gap

- Students with darker skin tones or atypical lighting are flagged at disproportionately higher rates
- Disabled and neurodivergent students are penalized for normal behavior — fidgeting, momentary movement, looking away
- As institutions move exams online permanently, building genuinely fair monitoring systems has become an urgent, practical need

Problem

Existing AI proctoring systems apply a single, population-wide detection standard to every student

- **Bias against darker-skinned students :**lower detection confidence → higher false-flag rates
- **Unfair to disabled/neurodivergent students :** normal behavior (looking away, stimming) gets penalized
- **Over-reliance on gaze alone:**a single, noisy signal used as the main detection basis
- **No human review:**many systems auto-penalize with no way for the student to contest a flag
- **Research gap**: no existing system is browser-based, privacy-preserving, individually-calibrated, and keeps the teacher in full control of the final decision

OBJECTIVE

1. **Real-Time Detection Track** gaze direction, tab-switching, focus loss, and copy-paste activity directly in the browser, with no server-side video processing

2. **Fairness Audit, Not Fairness Claims** Test MediaPipe against a second, independently trained model (RetinaFace) on FairFace to check if the same skin-tone confidence gap shows up in both offline, measured, honest.

3. **Personalized Fairness Calibrate** each student to their own behavioral baseline in the first 20-30 seconds, instead of one fixed rule for everyone

4. **Accommodation-Aware Scoring** Give disabled students shifted risk thresholds within the same scoring formula, not a separate system

5. Transparent Risk Output Produce an explainable green/yellow/red risk score showing exactly which signal contributed to a flag

6. **Human-in-the-Loop Control** Ensure the system only detects and informs; every consequence remains a decision made by the teacher

LITERATURE SURVEY

Paper	Authors / Year	Methodology	Dataset	Addresses	Limitation / Gap
AI-assisted Gaze Detection for Proctoring Online Exams	Shih et al., 2024	AI-based gaze detection; human–ML comparison	300 exam sessions, 11 proctors	Gaze-based suspicious behavior detection	Gaze-focused; no personalization or multi-signal risk scoring
Proctoring Online Exam Using Eye Tracking	Yaqub et al., 2023	Eye tracking + face hiding	5 exam/cheating videos	Privacy-preserving gaze monitoring	Very small dataset; mainly gaze-based
Examining Monitoring System: Detecting Abnormal Behavior	Ngo et al., 2024	Real-time abnormal behavior detection	Custom exam dataset	Real-time cheating/abnormal behavior detection	Predefined behaviors; limited personalization
Automated Online Exam Proctoring	Atoum et al., 2017	Multimodal: gaze, speech, phone & window detection	OEP dataset; 24 subjects	Multi-modal automated proctoring	Hardware requirements; privacy concerns
Detection of Malpractice by Head Pose & Gaze	Indi et al., 2021	Head-pose + eye-gaze estimation	10 volunteers	Off-screen attention detection	Small dataset; fixed thresholds
Cheating Detection through Eye Gaze Tracking	Dilini et al., 2021	WebGazer + One-Class SVM	5 undergraduate students	Browser-based gaze anomaly detection	Gaze-only; calibration/environment
Behavioral Patterns in Online Examination	Ferdosi et al., 2023	Facial landmarks + head/eye/lip orientation + K-NN	16 student volunteers	Behavioral/cheating score classification	Controlled data; limited personalization

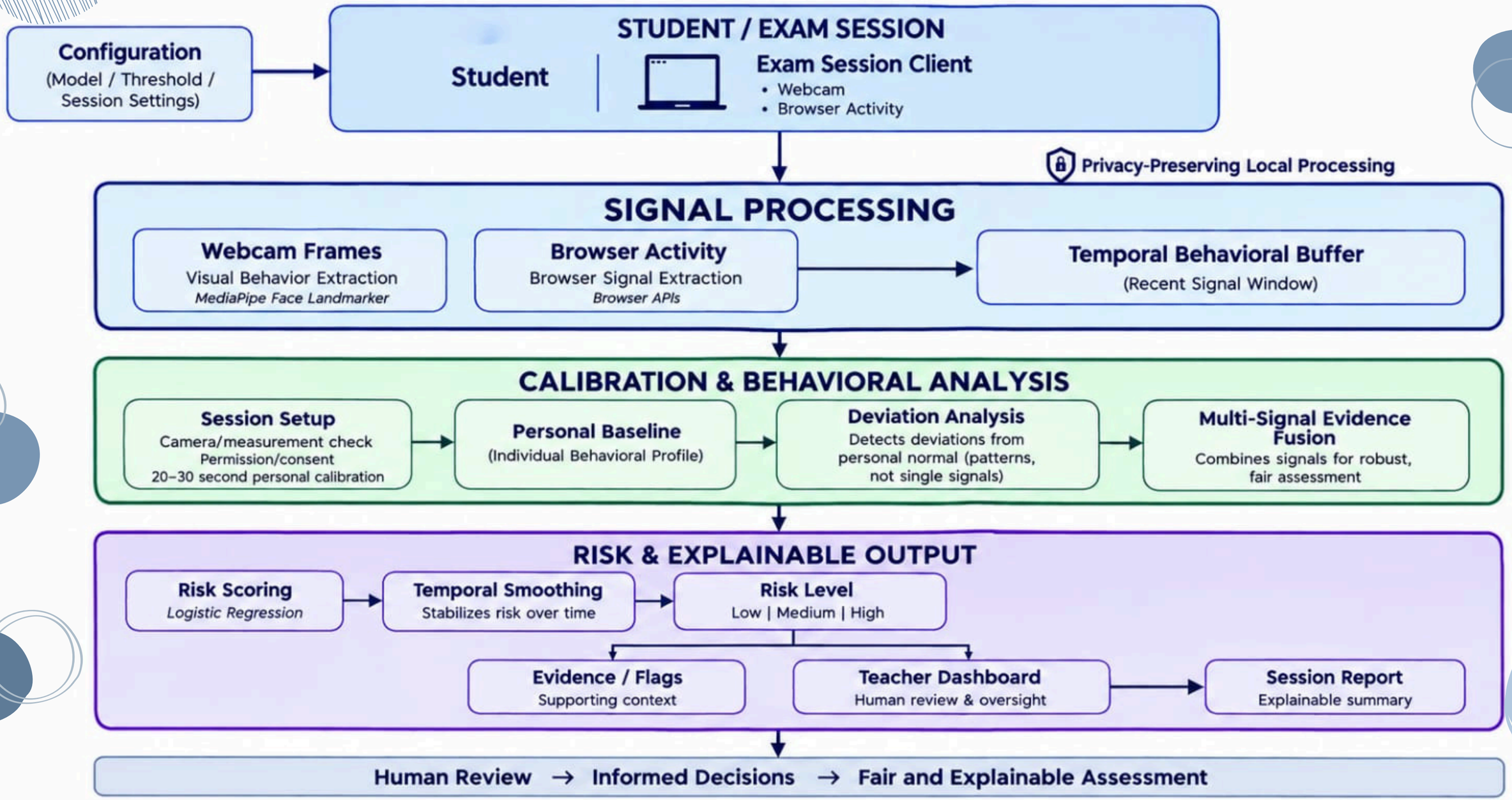
BASE PAPER

What we extend:
Add per-student baseline calibration — their system uses no personalization
Fine-tune RetinaFace on diverse skin-tone data for bias-aware detection
Fuse gaze with tab-switch, focus-loss, and copy-paste signals
Add accommodation-aware thresholds and measured fairness via FairFace

Why this paper:
2024 publication, directly focused on gaze-based exam proctoring
Industry system (Duolingo) built to assist human proctors, not replace them
Freely accessible on arXiv

AI-assisted Gaze Detection for Proctoring Online Exams"
Yong-Siang Shih, Zach Zhao, Chenhao Niu, Bruce Iberg, James Sharpnack, Mirza Basim Baig — Duolingo, Inc.
arXiv:2409.16923 • 2024

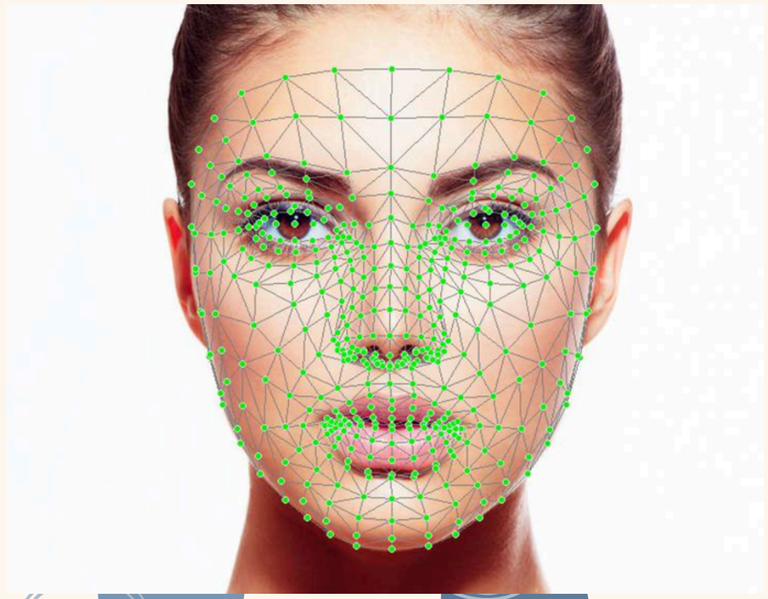
GlassBox: Personalized Calibration for Privacy-Preserving and Explainable Risk Scoring in Online Exam Proctoring



MODULES

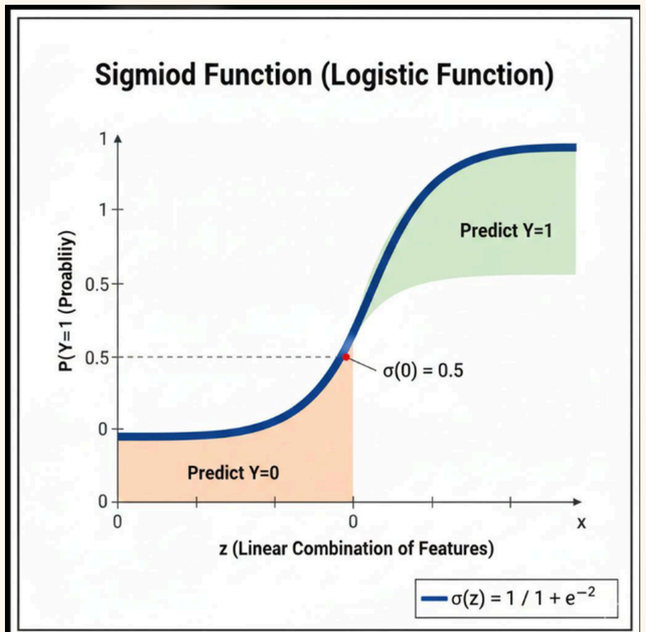
1. *MediaPipe-Face and movement tracking*

It helps detect facial landmarks and estimate observable signals, such as head position and eye-related movement, from webcam input.



2. *Logistic Regression-Risk prediction*

It uses input signals to estimate a prediction score. In GlassBox, the proposed use is to combine behavioral indicators into an interpretable risk estimate.



Module Project Implementation / Prototype

GLASSBOX Home Monitor Sessions Demo user: Reviewer Sign out

GLASSBOX
Explainable behaviour monitoring

BACKEND Connected · TRACKER Ready · BASELINE Calibrated · Monitoring active

Camera	MediaPipe	Feature extraction	Behaviour analysis	GlassBox explanation	Dashboard
Live	Face landmarker ready	1773 signals	2 events	Risk low	Live

Live monitoring — Recording



Calibration complete. Personal baseline established. Recalibrate

WHY WAS THIS FLAGGED? **10** /100 LOW

Session #1

Collection condition: Normal

Look at the screen normally and behave naturally.

End session

Status	Session active	Elapsed	00:32
Signals	1773	Events	2
Session ID	3		

Behaviour indicators — Latest reading

✓ Face detection — Landmarks tracked	Detected	100% quality
! Gaze — Deviation from baseline	Notable	1.5 σ
✓ Head movement — Yaw / pitch / roll offset	Within baseline	0° 5° 4°
✓ Window focus — Browser window / tab	In focus	–
– Posture	Not available	

Session history

New session

Recorded monitoring sessions on this machine. The risk score is a rule-based evidence summary, not a probability of cheating.

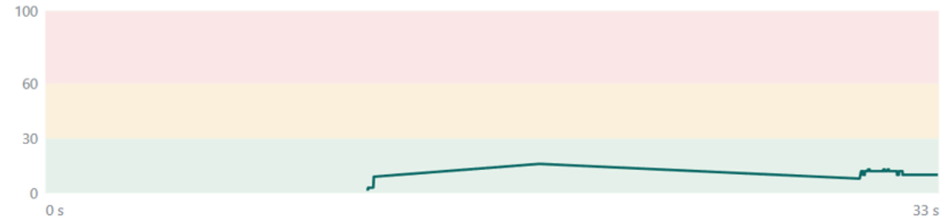
SESSION	STARTED	DURATION	CONDITION	STATUS	SIGNALS	EVENTS	PEAK RISK	
#3	10/21/2026, 10:50:52 PM	0:33	NORMAL	COMPLETED	83	2	16 · LOW	Details →
#2	10/21/2026, 10:49:12 PM	0:33	NORMAL	COMPLETED	85	1	11 · LOW	Details →
#1	10/21/2026, 10:41:28 PM	0:25	NORMAL	COMPLETED	75	0	—	Details →

Session #3

10/21/2026, 10:50:52 PM · COMPLETED · condition **NORMAL**

DURATION	EVIDENCE EVENTS	PEAK RISK	FINAL RISK
0:33	2	16 LOW	10

Risk timeline



Risk score over time (bands: LOW < 30 ≤ MEDIUM < 60 ≤ HIGH). A rule-based summary, not a probability.

Evidence events

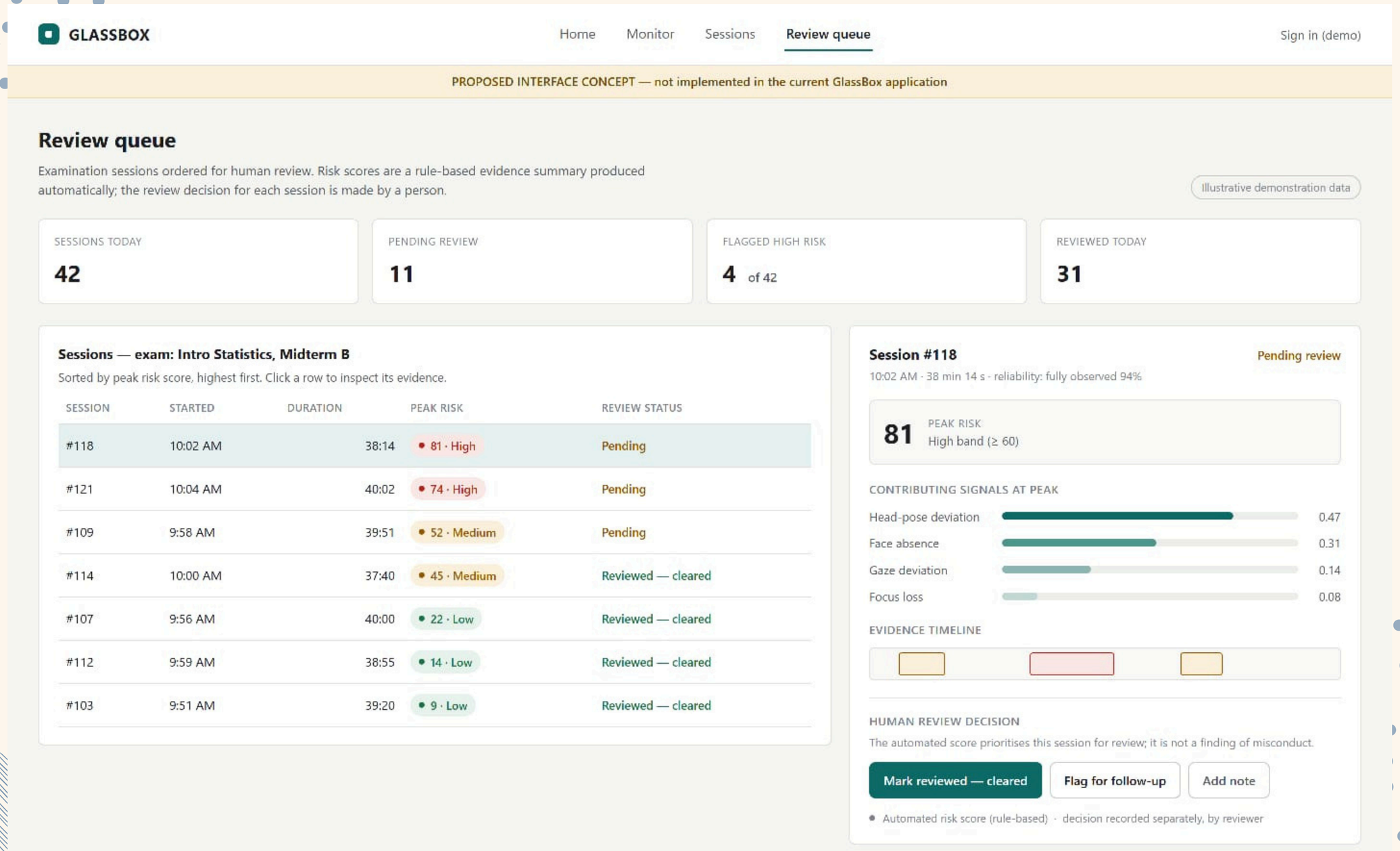
AT	EVENT	DURATION	STRENGTH
+18.1 s	Focus loss	6.1 s	1.00
+29.9 s	Face absence	72 ms	0.01

Recorded signals

Counts of persisted raw signals (vision samples are stored about once per second).

FACE_QUALITY 26 · GAZE_DEVIATION 26 · HEAD_POSE_DEVIATION 26 · FOCUS_LOST 1 · TAB_HIDDEN 1 · TAB_VISIBLE 1 · FOCUS_GAINED 1 · FACE_ABSENT 1

Module Project Implementation / Prototype



REFERENCES

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[2] E. Heinrich, "A Systematic-Narrative Review of Online Proctoring Systems and a Case for Open Standards," Open Praxis, vol. 17, no. 3, pp. 485–499, 2025.

[3] D. R. Yoder-Himes et al., "Racial, skin tone, and sex disparities in automated proctoring software," Frontiers in Education, vol. 7, 2022.

[4] W. Belzak, J. Burstein, and A. A. von Davier, "Evaluating Fairness in AI-Assisted Remote Proctoring," Proceedings of Machine Learning Research, vol. 273, pp. 125–132, 2025.

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GANTT CHART

Planned Project Timeline (July – October)

